

Western New England University
College of Engineering
ECE Department
Wave Transmission and Reception
EE 457/557
Fall 2025
Design Project #3
Due: November 17, 2025

Name: Steven Placzek

References: Ryan Leonard

Tommy Nguyen

Kevin Farjado

Aidan Butler

--

Design Project #3

	Score	Max
Design		200
HFSS		300
Tables		100
Presentation		100
Total		700

Design Project #3

3-GHz Coaxial-Fed Microstrip Patch Antenna

Steven Placzek

EE-457 Wave Transmission and Reception
Western New England University

EE-457

Wave Tx and Rx
Design Project 03

Table of Contents

- ① Project Overview
 - Project Objectives, Specifications, Initial Parameters, & HFSS Setup
- ② Table 1
 - Table 1: Microstrip Patch Design Parameters Calculated in Matlab ©
- ③ Table 2
 - Table 2: Geometry Parameters per HFSS Optimetric Sweep
- ④ Table 3
 - Table 3: HFSS Antenna Parameters
- ⑤ BW_f
 - Table 4: 10 dB BW_f Summary
 - Sweep Bandwidths
- ⑥ Final Microstrip Patch Design: Antenna Parameters
 - Final HFSS Antenna Parameters
 - Microstrip Patch Antenna Design Gain (V4)
 - HFSS Simulated Radiation Pattern
 - Microstrip Patch Antenna Design Impedance Match
 - BW_f at & -10 dB Points
- ⑦ Conclusion
 - Summary
- ⑧ Appendices
 - Table of Appendices



Project Objectives and Specifications

Design Objectives

- Design a **3-GHz** coaxial-fed microstrip patch antenna on FR4 ($h = 1.5748$ mm, $\epsilon_r = 4.4$, $\tan \delta = 0.02$)
- Set patch width target $W \approx 1.5 \cdot L_e$
- Meet input return loss at 3 GHz: $|S_{11}| \leq -30$ dB
- Characterize gain, directivity, E/H beam-widths, and radiation efficiency

Table 1



Parameter	Calculated	Unit
f_0	3.0000	GHz
h	1.5748	mm
ϵ_r	4.4	–
λ_g	47.6401	mm
L_e	23.8201	mm
W'	35.7301	mm
$\epsilon_{r,\text{eff}}$	4.4000	–
ΔL	726.7002	μm
L_p	22.3667	mm
W	35.7301	mm
R_{Edge}	239.4675	Ω
x_f	8.3119	mm

Table 1: Calculated design parameters for the 3-GHz coaxial-fed microstrip patch antenna

Table 2



Parameter	V1	V2	V3	V4	Unit
L_e	23.8201	23.8201	23.8201	24.0187	mm
ΔL	726.7001	726.7001	726.7001	726.7001	μm
L	22.3667	22.3667	22.3667	22.5652	mm
W	35.7302	35.7302	35.7302	36.0279	mm
x_f	8.3119	8.3119	8.3119	5.8183	mm
x_0	3.5982	3.5982	3.5982	6.1910	mm
L_s	33.5501	33.5501	33.5501	33.8478	mm
W_s	53.5952	53.5952	53.5952	54.0418	mm

Table 2: Simulated parameters for HFSS optimetric sweep of X_{sweep} . Note: V2 and V3 did not see any improvements for $|S_{11}|_{\text{dB}}$ after running another sweep, despite changing the step size to 0.02.

Table 3



Parameter	HFSS	Unit
D	5.09	W/W
D	7.07	dB
G	2.42	W/W
G	3.84	dB
BW_{EP}	102.94	°
BW_{HP}	78.68	°
e_{rad}	47.5	%

Table 3: Simulated directivity, gain, beam-width, and radiation efficiency at 3 GHz

Table 4



Parameter	Calculated	HFSS	Unit
$BW_f (-10 \text{ dB})$	21.6673	89.7000	MHz
$BW_f (-10 \text{ dB})$	0.7222	2.9898	%

Table 4: Calculated and simulated -10 dB impedance bandwidth

Key Finding

Achieved approximately 3% **fractional bandwidth** at the -10 dB return loss level for optimized designs (V2–V4).



Bandwidth Summary by Sweep

Parameter	Unit	V1	V2	V3	V4
$BW_f (-10 \text{ dB})$	GHz	N/A	0.0903	0.0903	0.0897
$BW_f (-10 \text{ dB})$	%	N/A	2.99	2.99	2.99

Table 5: -10 dB bandwidths from $|S_{11}|_{(\text{dB})}$ for sweeps V1–V4



Quantity	Value	Unit
Frequency f	3.0000	GHz
Max U	0.3850	W/sr
Peak directivity	4.3989	—
Peak gain	2.4211	—
Peak realized gain	2.4192	—
Radiated power	1.0999	W
Accepted power	1.9984	W
Radiation efficiency	0.5504	—
Total efficiency	0.5500	—
Front-to-back ratio	6.1402	dB

Table 6: HFSS antenna parameters at $f_0 = 3.0$ GHz for final microstrip patch antenna design.

Gain Plot

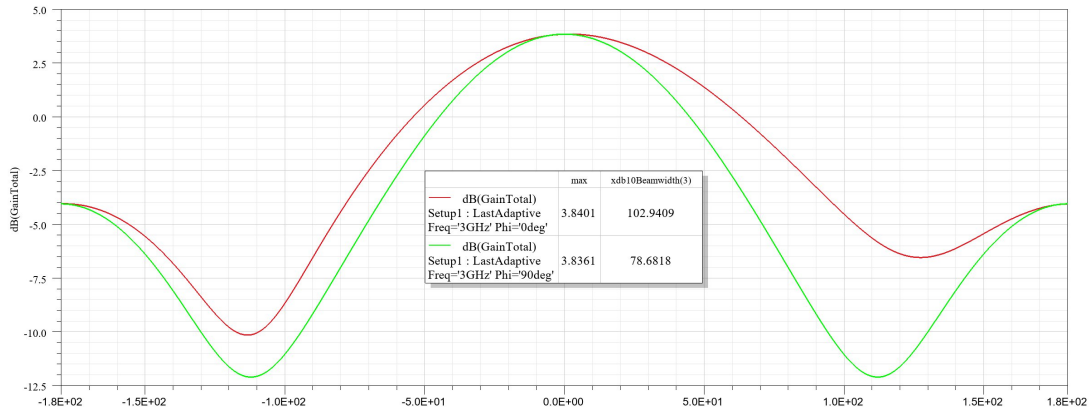


Figure 1: Total gain patterns at 3 GHz for the final optimized design, highlighting the 10 dB beam-width in the principal planes.

Radiation Pattern

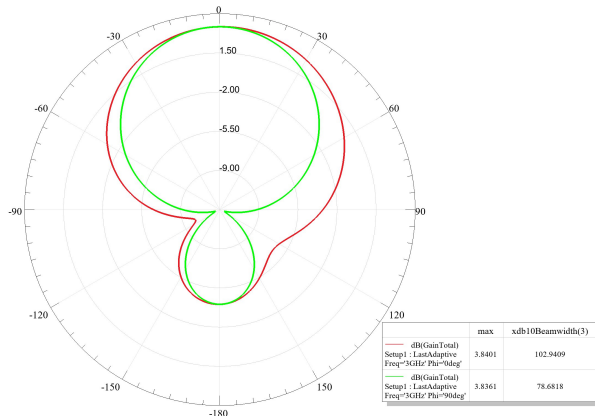


Figure 2: HFSS Simulated Radiation Pattern with E-plane & H-plane 10 dB beam-widths.

Smith Chart

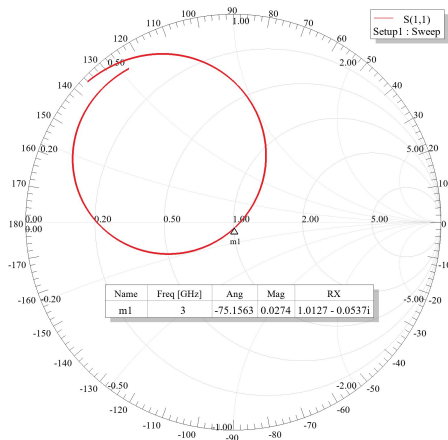


Figure 3: $|S_{11}|_{(\text{dB})}$ design impedance from 2.5 to 3.5 GHz: demonstrates a good match at 3 GHz

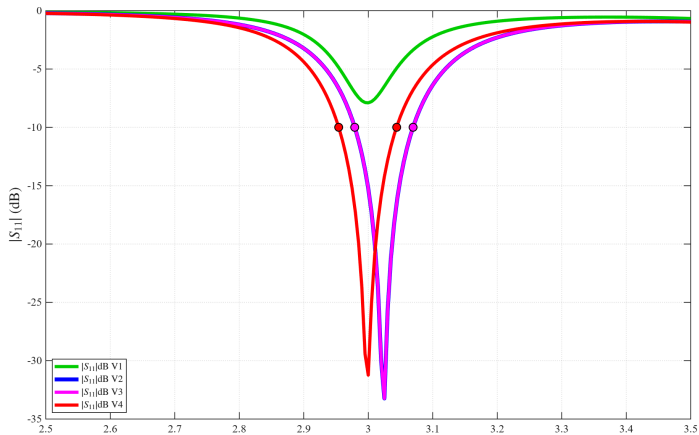


Figure 4: $|S_{11}|$ (dB) vs f (GHz) — MATLAB plot of the -10 dB points, per optimetric sweep.



Summary & Conclusions

Accomplishments:

- ① Successfully designed a 3-GHz coaxial-fed microstrip patch antenna
- ② Optimized feed position through parametric HFSS sweeps
- ③ Achieved target $|S_{11}| \leq -30$ dB at resonance
- ④ Characterized E-plane and H-plane radiation patterns

Key Results:

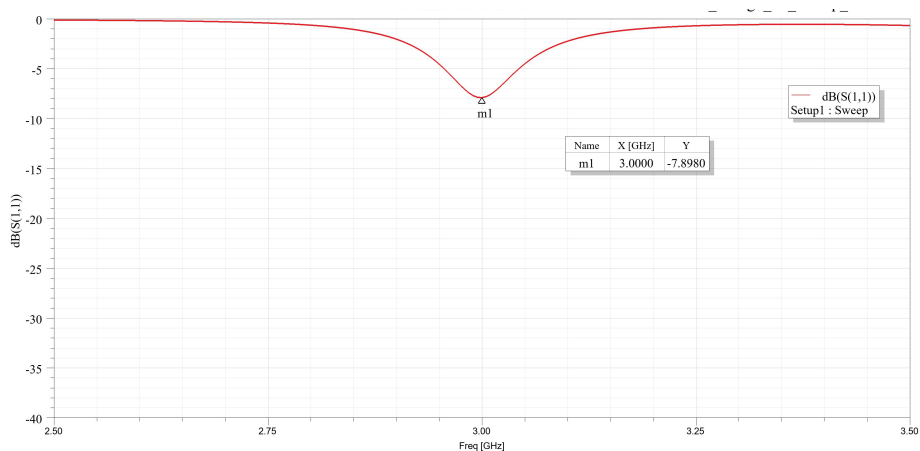
- Fractional bandwidth (-10 dB): $\sim 3\%$
- Final optimized dimensions validated through simulation
- Good impedance match demonstrated on Smith chart



Appendices

⑧ Appendices

■ Table of Appendices

Figure 5: Optimetric Sweep V1: $|S_{11}|_{(\text{dB})}$ vs $f_{(\text{GHz})}$ (initial design).

EE-457

Wave Tx and Rx
Design Project 03

V2 (All Traces)

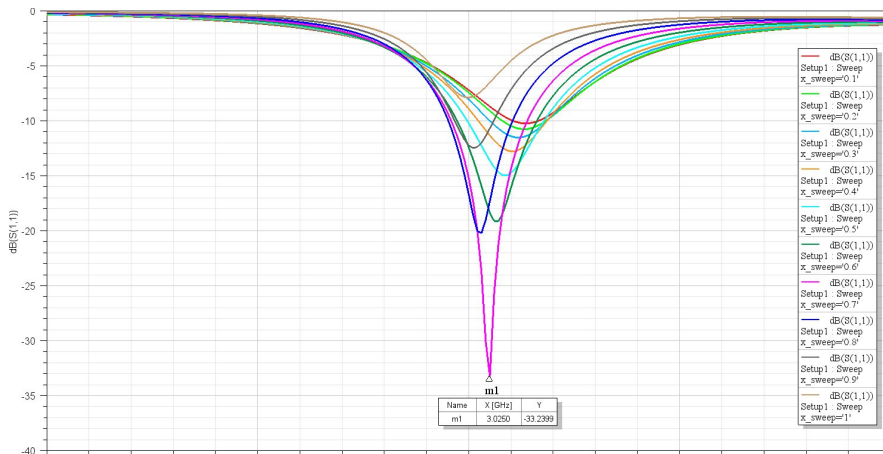


Figure 6: V2: All HFSS optimetric iterations for X_{sweep} . Linear frequency sweep from 2.5 GHz to 3.5 GHz with step $\Delta f = X_{sweep}/10$, spanning $X_{sweep}/10$ to X_{sweep} .

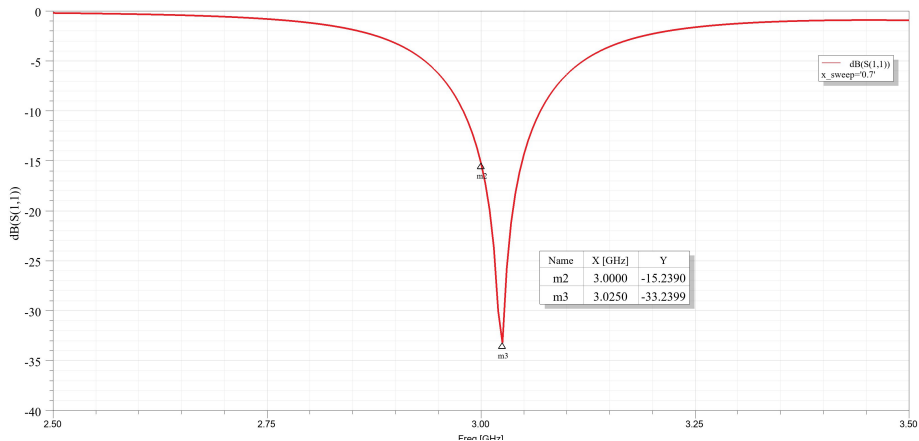


Figure 7: HFSS Optimetric Sweep 02: Best result from optimetric Sweep 02

Sweep 03

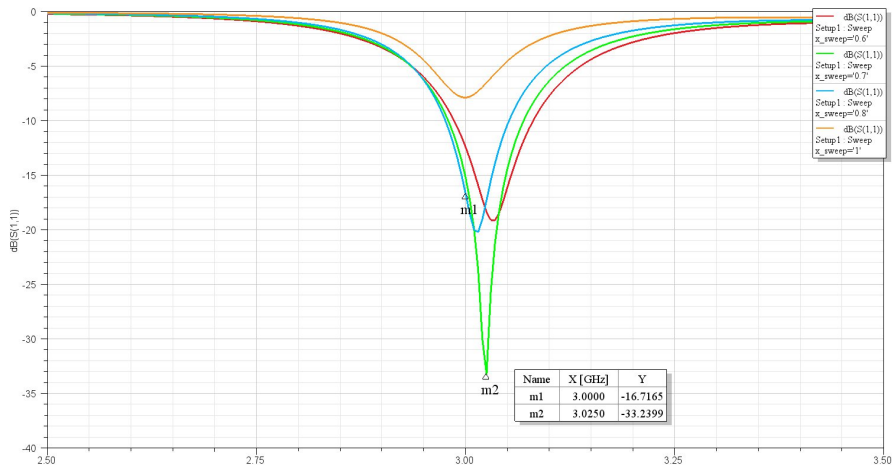


Figure 8: S-Parameters: Sweep 03_{x_{sweep}} with 0.02 GHz step refinement

Sweep 03 (Refined)

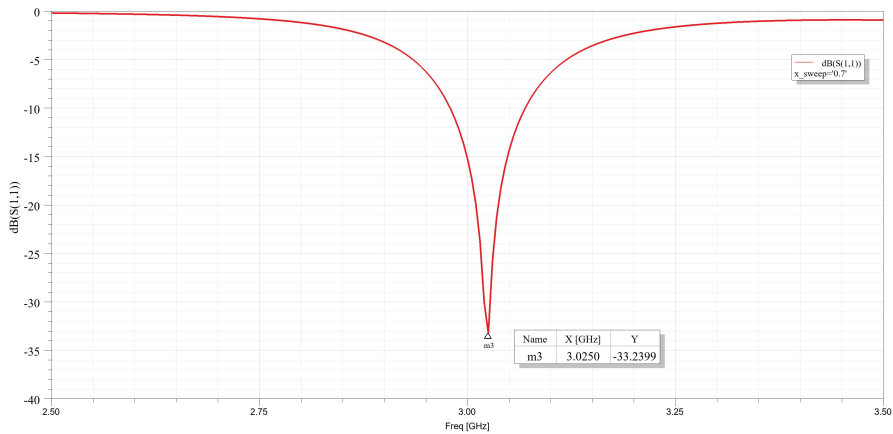


Figure 9: S-Parameters: Sweep 03 (Refined) Refined sweep focused near the resonance

Sweep 04

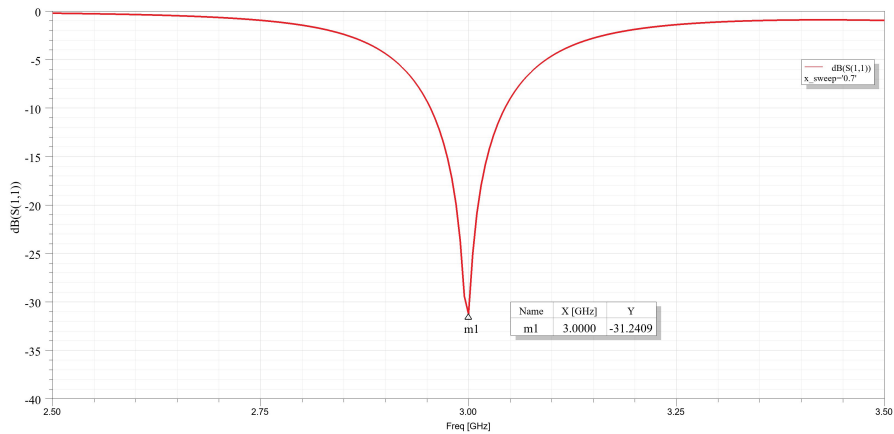


Figure 10: S-Parameters: Sweep 04 Final optimized sweep for $|S_{11}|$

EE-457

Wave Tx and Rx
Design Project 03

Appendix I: MATLAB Design Code

```

1  %% -----
2  % EE_457_Design_Project_03_Placzek_Matlab.m
3  % Calculations and Design for EE-457 Design Project 03.
4  % Implements the transmission-line patch calculations for EE-457.
5  % IMPORTANT NOTE: This script relies on the presence of Dr.
6  %   Burke's
7  % MATLAB_EELab/* MATLAB scripts in the MATLAB PATH in order to
8  %   run.
9  % (e.g., EE457_Microstrip_Patch_Conductance).
10
11 clc; close all; clear all;
12
13 projDir = fileparts(mfilename('fullpath'));
14 rootDir = fileparts(projDir);
15 eelabRoot = fullfile(rootDir, 'MATLAB_EELab');
16 % if exist(eelabRoot, 'dir') && ~contains(path, eelabRoot)
17 %     addpath(genpath(eelabRoot));
18 % else
19 %     warning('MATLAB_EELab path not found: %s', eelabRoot);
20 % end
21
22 %% -----
23 % Input specifications

```

```

48  %% -----
49  % DEFAULTSPECS --> Project inputs and physical constants.
50
51 function specs = defaultSpecs()
52 specs.f0 = 3e9;           % Hz
53 specs.h = 1.5748e-3;      % m (62 mil)
54 specs.er = 4.4;          % FR-4  $\epsilon_r$ 
55 specs.tand = 0.02;        % Loss tangent (documentation)
56 specs.t_patch = 18e-6;    % Copper thickness (m)
57 specs.rin = 50;           % Target input resistance (ohm)
58 specs.widthFactor = 1.5;  % W ~ 1.5 * Le (assignment
59 %   requirement)
60 specs.substrateFactor = 1.5; % Ls ~ Ws ~ 1.5 * patch size
61 specs.c0 = 299792458;
62 specs.lambda0 = specs.c0/specs.f0;
63 specs.k0 = 2*pi/specs.lambda0;
64 specs.eta0 = 376.730313668;
65 end
66
67 %% -----
68 % computeDimensions --> Does exactly what it sounds like it
69 %   would do.

```

EE-457

Wave Tx and Rx
Design Project 03

Appendix I: MATLAB Design Code (cont.)

```

102 %%-----
103 % FRINGINGEXTENSION -> Hammerstad extension length on one side.
104 function deltaL = fringingExtension(specs, W, eps_eff)
105 ratio = W/specs.h;
106 deltaL = 0.412*specs.h * ((eps_eff + 0.3)*(ratio +
107     0.264))/((eps_eff - 0.258)*(ratio + 0.8));
108 end
109 %% COMPUTEEDGEQUANTITIES -> Calls the EE457 conductance helper
110     script.
111 function edge = computeEdgeQuantities(specs, dims)
112 [Gedge, G1, G12, B1, G1a, B1a] =
113     EE457_Microstrip_Patch_Conductance( ...
114     dims.Le, dims.W, specs.h, specs.k0, specs.lambda0,
115     specs.eta0);
116 edge.Gedge = Gedge;
117 edge.G1 = G1;
118 edge.G12 = G12;
119 edge.B1 = B1;
120 edge.G1_approx = G1a;
121 edge.B1_approx = B1a;
122 edge.Redge = 1/Gedge;
123 end

```

```

148 %% -----
149 % ESTIMATEBANDWIDTH --> -10 dB impedance bandwidth
150     (transmission-line Q).
151 function bw = estimateBandwidth(specs, dims)
152 er = specs.er;
153 h = specs.h;
154 lambda0 = specs.lambda0;
155 fracBW = 3.77 * ((er - 1)/(er^2)) * (h/lambda0);
156 VSWR = 1.92; % -10 dB return-loss circle
157 fracBW = fracBW * sqrt((VSWR - 1)/VSWR);
158 bw.abs = fracBW * specs.f0;
159 bw.frac = fracBW;
160 bw.percent = fracBW * 100;
161 end
162 %% -----
163 % ASSEMBLERESULTS --> Organizes script output values for
164     console and CSV.
165 function results = assembleResults(specs, dims, edge, feed,
166     pattern, bw)
167 mm = 1e-3;

```

EE-457

Wave Tx and Rx
Design Project 03

Appendix I: MATLAB Design Code (cont.)

```

193 %% -----
194 % DISPLAYSUMMARY --> Print concise summary for sanity checks.
195
196 function displaySummary(results)
197 fprintf('--- EE-457 Design Project 03: Table 01 Calculations --
198 \n');
199 fprintf('\nf0 = %.4f GHz\n\ner = %.4f\n\nh = %.4f mm\n',
200 results.f0_GHz, results.er, results.h_mm);
201 fprintf('\nlg = %.4f mm\n\nLe = %.4f mm\n\nW = %.4f mm\n\nL =
202 %.4f mm\n', ...
203 results.lambda_g_mm, results.Le_mm, results.W_mm,
204 results.L_mm);
205 fprintf('\n\DeltaL = %.4f \mu m\n\nR_edge = %.4f \Omega\n\nxf = %.4f mm (from
206 edge)\n', ...
207 results.deltaL_um, results.Redge_ohm, results.xf_mm);
208 fprintf('\nHPBW_E = %.4f^\n\nHPBW_H = %.4f^\n\nD \approx %.4f (%.4f
209 dB)\n', ...
210 results.BWE_deg, results.BWH_deg, results.D_lin,
211 results.D_dB);
212 fprintf('\nEstimated BW10dB = %.4f MHz (%.4f%%)\n',
213 results.BW_MHz, results.BW_percent);
214 end

```

```

207 %% -----
208 % WRITETABLES --> Populate calculated columns for Tables 1 and
209 4.
210
211 function writeTables(results)
212 rootDir = fileparts(mfilename('fullpath'));
213 tablesDir = fullfile(rootDir, 'tables');
214 if ~exist(tablesDir, 'dir')
215     mkdir(tablesDir);
216 end
217 T1 = table(results.f0_GHz, results.h_mm, results.er,
218 results.lambda_g_mm, ...
219 results.Le_mm, results.Wprime_mm, results.eps_eff,
220 results.deltaL_um, ...
221 results.L_mm, results.W_mm, results.Redge_ohm,
222 results.xf_mm, ...
223 'VariableNames',
224 {'f0_GHz', 'h_mm', 'eps_r', 'lambda_g_mm', 'Le_mm', 'Wprime_mm',
225 'eps_eff', 'deltaL_um', 'L_mm', 'W_mm', 'R_edge_ohm', 'xf_mm'});
226 writetable(T1, fullfile(tablesDir, 'table_1_calculated.csv'));
227 T4 = table(results.BW_MHz, results.BW_percent, ...
228 'VariableNames', {'BW_10dB_MHz', 'BW_10dB_percent'});
229 writetable(T4, fullfile(tablesDir, 'table_4_calculated.csv'));

```



EE-457
Wave Tx and Rx
Design Project 03

Appendix II: MATLAB –10 dB Bandwidth Code

```

1  % EE_457_Design_Project_03_Placzek_Matlab_read_csv_10dB.m
2  % Matlab script for creating |S11|dB versus frequency plots
3  % Uses all *.csv files in the current project directory.
4  % Assumptions (HFSS-style export):
5  %   - Each CSV file has at least three columns:
6  %       Col 1: sweep variable (ignored here)
7  %       Col 2: frequency (GHz by default; see PlotSpec.freqInHz)
8  %       Col 3: |S11| in dB
9  %   - Files are named so that alphabetical sort produces V1, V2,
10 %      V3, V4, ...
11 % Behavior:
12 %   - All traces are plotted on a single figure.
13 %   - All traces get -10 dB markers at crossings.
14 %   - A -10 dB bandwidth is computed and printed for each trace
15 %     (BWf),
16 %     or reported as N/A if there are fewer than two -10 dB
17 %     crossings.
18 %
19 % -----
20 %
21 % Plot configuration and file discovery

```

```

30 %% -----
31 %% DEFAULT_PLOT_SPEC --> Styles and options for S11 plotting.
32
33 function PlotSpec = default_Plot_Spec()
34 PlotSpec = struct();
35 PlotSpec.colorList = [
36     0.0  0.8  0.0; % V1 - green
37     0.0  0.0  1.0; % V2 - blue
38     1.0  0.0  1.0; % V3 - magenta
39     1.0  0.0  0.0; % V4 - red
40 ];
41 PlotSpec.lineWidths = [4 5 4 4];
42 PlotSpec.markerLevel_dB = -10; % Level for bandwidth
43 % markers (dB)
44 PlotSpec.markerIndices = []; % [] => mark all traces
45 PlotSpec.lineWidth = 3; % fallback if lineWidths
46 % not used
47 PlotSpec.markerSize = 10;
48 PlotSpec.fontSizeAxes = 14;
49 PlotSpec.fontSizeLabel = 18;
50 PlotSpec.fontSizeLegend = 12;
51 PlotSpec.fontName = 'Times New Roman';
52 PlotSpec.gridLineStyle = ':';
53 PlotSpec.gridLineWidth = 1;

```



Appendix II: MATLAB –10 dB Bandwidth Code (cont.)

```

72 %% -----
73 % PLOT_ALL_S11 --> Read CSVs, plot all |S11| curves, and
    decorate axes.
74
75 function Fig = plot_All_S11(files, PlotSpec)
76 Fig = figure; hold on; grid on; box on;
77 set(Fig, 'Color', 'w');
78 nFiles = numel(files);
79 nColors = size(PlotSpec.colorList, 1);
80 fprintf('--- %.1f dB bandwidths from |S_{11}| ---\n', ...
81         abs(PlotSpec.markerLevel_dB));
82 for k = 1:nFiles
83     filePath = fullfile(files(k).folder, files(k).name);
84     [f, s11] = read_S11_Trace(filePath, PlotSpec.freqInHz);
85     if isempty(f) || numel(s11) < 2
86         warning('Skipping file with insufficient data: %s',
87                 files(k).name);
88         continue;
89     end
90     c = PlotSpec.colorList(min(k, nColors), :);
91     name = strip_Extension(files(k).name);
92     if isfield(PlotSpec, 'lineWidths') &&
        ~isempty(PlotSpec.lineWidths)
        lwList = PlotSpec.lineWidths;
    end

```

```

126 %% -----
127 % READ_S11_TRACE --> Load frequency and |S11| from a CSV file.
128
129 function [f, s11] = read_S11_Trace(filePath, freqInHz)
130 T = readtable(filePath, 'PreserveVariableNames', true);
131 vars = T.Properties.VariableNames;
132 freqIdx = find(contains(vars, 'Freq', 'IgnoreCase', true), 1,
133                'first');
134 s11Idx = find(contains(vars, 'dB', 'IgnoreCase', true) | ...
135               contains(vars, 'S11', 'IgnoreCase', true), 1,
136               'first');
137 if isempty(freqIdx) || isempty(s11Idx)
138     warning('Could not find frequency/S11 columns in %s.
139             Skipping.', filePath);
140     f = [];
141     s11 = [];
142     return;
143 end
144 f = T{:, freqIdx};
145 s11 = T{:, s11Idx};
146 % Extract numeric data
147 bad = ~isfinite(f) | ~isfinite(s11);
148 % Drop any NaNs
149 f(bad) = [];
150 s11(bad) = [];
151 if numel(f) < 2 || numel(s11) < 2
152     % Still not enough

```

EE-457

Wave Tx and Rx
Design Project 03

Appendix II: MATLAB –10 dB Code (cont.)

```

175 %% -----
176 % COMPUTEBANDWIDTHFROMLEVEL --> -10dB bandwidth around the
    resonance.
177
178 function bw = computeBandwidthFromLevel(f, s11, level_dB)
179 bw = struct(); % Initialize
    struct with NaNs
180 bw.f0 = NaN;
181 bw.fLow = NaN;
182 bw.fHigh = NaN;
183 bw.BWabs = NaN;
184 bw.BWfrac = NaN;
185 bw.BWpercent = NaN;
186 bw.crossings = [];
187 if numel(f) < 2 || numel(s11) < 2
188     return;
189 end
190 crossings = levelCrossings(f, s11, level_dB);
191 if numel(crossings) < 2
192     return;
193 end
194 [~, idxMin] = min(s11);
195 f0 = f(idxMin);
196 fLow = crossings(1);

```

```

209 %% -----
210 % FORMAT_S11_AXES --> Publication-style axis, labels, and
    legend.
211
212 function formatS11Axes(PlotSpec)
213 set(gca, ...
214     'FontSize', PlotSpec.fontSizeAxes, ...
215     'FontName', PlotSpec.fontName, ...
216     'GridLineStyle', PlotSpec.gridLineStyle, ...
217     'LineWidth', PlotSpec.gridLineWidth);
218 xlabel('$f$ (GHz)', 'Interpreter', 'latex', ...
219     'FontSize', PlotSpec.fontSizeLabel);
220 ylabel('$|S_{11}|$ (dB)', 'Interpreter', 'latex', ...
221     'FontSize', PlotSpec.fontSizeLabel);
222 title(PlotSpec.titleString, 'Interpreter', 'latex', ... % Title
223     'FontSize', PlotSpec.fontSizeLabel);
224 if isfield(PlotSpec, 'legendStrings') &&
    ~isempty(PlotSpec.legendStrings)
225     legend(PlotSpec.legendStrings, ...
226         'Interpreter', 'latex', ...
227         'Location', 'southwest', ...
228         'FontSize', PlotSpec.fontSizeLegend);
229 else
230     legend('show', 'Location', 'southwest', ...

```

EE-457

Wave Tx and Rx
Design Project 03

Appendix III: HFSS Parameter Values

Name	Value	Unit	Evaluated V...	Type
h	1.5748	mm	1.5748mm	Design
Le	23.8201	mm	23.8201mm	Design
del_L	726.7001	um	726.7001um	Design
t	18	um	18um	Design
xf	8.3119	mm	8.3119mm	Design
L	$Le - 2 * del_L$		22.3667mm	Design
W	$01.5 * Le$		35.73015mm	Design
LS	$1.5 * L$		33.55005mm	Design
WS	$1.5 * W$		53.595225mm	Design
x0	$(1/2) * Le - xf$		3.59815mm	Design
ri	0.635	mm	0.635mm	Design
LC	17	mm	17mm	Design
ro	2.032	mm	2.032mm	Design
ro2	$ri + ro$		2.667mm	Design
LC2	$LC + (1mm)$		18mm	Design

Figure 11: Sweep 00 (Initial)

EE-457

Wave Tx and Rx
Design Project 03

Appendix III: HFSS Parameter Values (cont.)

Name	Value	Unit	Evaluated V...	Type
h	1.5748	mm	1.5748mm	Design
Le	23.8201	mm	23.8201mm	Design
del_L	726.7001	um	726.7001um	Design
t	18	um	18um	Design
xf	8.3119*x_sweep	mm	8.3119mm	Design
L	Le - 2*del_L		22.3667mm	Design
W	01.5 * Le		35.73015mm	Design
LS	1.5 * L		33.55005mm	Design
WS	1.5 * W		53.595225mm	Design
x0	(1/2) * Le - xf		3.59815mm	Design
ri	0.635	mm	0.635mm	Design
LC	17	mm	17mm	Design
ro	2.032	mm	2.032mm	Design
ro2	ri + ro		2.667mm	Design
LC2	LC + (1mm)		18mm	Design
x_sweep	1		1	Design

Figure 12: Sweep 01

EE-457

Wave Tx and Rx
Design Project 03

Appendix III: HFSS Parameter Values (cont.)

Name	Value	Unit	Evaluated V...	Type
h	1.5748	mm	1.5748mm	Design
Le	23.8201	mm	23.8201mm	Design
del_L	726.7001	um	726.7001um	Design
t	18	um	18um	Design
xf	8.3119*x_sweep	mm	8.3119mm	Design
L	Le - 2*del_L		22.3667mm	Design
W	01.5 * Le		35.73015mm	Design
LS	1.5 * L		33.55005mm	Design
WS	1.5 * W		53.595225mm	Design
x0	(1/2) * Le - xf		3.59815mm	Design
ri	0.635	mm	0.635mm	Design
LC	17	mm	17mm	Design
ro	2.032	mm	2.032mm	Design
ro2	ri + ro		2.667mm	Design
LC2	LC + (1mm)		18mm	Design
x_sweep	1		1	Design

Figure 13: Sweep 03

EE-457

Wave Tx and Rx
Design Project 03

Appendix III: HFSS Parameter Values (cont.)

Name	Value	Unit	Evaluated V...	Type
h	1.5748	mm	1.5748mm	Design
Le	24.0186008	mm	24.018601mm	Design
del_L	726.7001	um	726.7001um	Design
t	18	um	18um	Design
xf	8.3119*x_sweep	mm	5.81833mm	Design
L	Le - 2*del_L		22.565201mm	Design
W	01.5 * Le		36.027901mm	Design
LS	1.5 * L		33.847801mm	Design
WS	1.5 * W		54.041852mm	Design
x0	(1/2) * Le - xf		6.1909704mm	Design
ri	0.635	mm	0.635mm	Design
LC	17	mm	17mm	Design
ro	2.032	mm	2.032mm	Design
ro2	ri + ro		2.667mm	Design
LC2	LC + (1mm)		18mm	Design
x_sweep	0.7		0.7	Design

Figure 14: Sweep 04

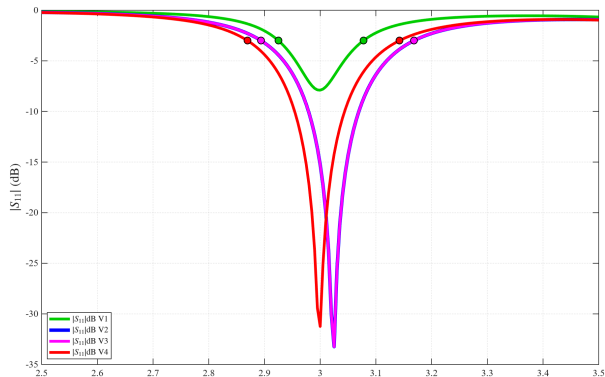
3 dB BW_f Plot

Figure 15: S_{11} (dB) vs f (GHz): MATLAB plot of the -3 dB points, per optimetric sweep in HFSS.



Appendix IV: References

References:

Ryan Leonard, Aidan Butler, Tommy Nguyen, Fevin Farjado

Software Tools:

- Ansys HFSS for electromagnetic simulation
- MATLAB for data analysis and visualization
- L^AT_EX for presentation formatting

Course:

EE-457/557 Wave Transmission and Reception

Fall 2025